



IILM UNIVERSITY, GREATER NOIDA

Land Degradation Prediction System

A machine-learning system for classifying land degradation severity across Uttar Pradesh

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Introduction

Why LULC and land degradation matter

Land Use Land Cover (LULC): Classifies the earth's surface into categories — forest, agriculture, built-up, water, and barren land — using satellite-derived indicators.

Land Degradation: The decline in land quality caused by deforestation, urban expansion, soil erosion, and unsustainable land use over time.

Why it matters: Identifying degraded land early helps prioritize restoration and supports better land-use policy decisions.

My approach: Instead of visual inspection, I use a trained machine learning model to classify degradation severity automatically from environmental data.

Study Area: Uttar Pradesh

Key Facts

State	Uttar Pradesh, India
Districts covered	20
Grid resolution	5 km × 5 km
Time period	2020–2024
Total observations	22,645 grid cells

Notes

- Districts include Chandauli, Badaun, Banda, Aligarh and others across the state
- Wide mix of cropland, forest, grassland, and built-up land across districts
- Diverse hydrology and soil-moisture conditions across the Gangetic plain

A state-wide grid gives the model enough variety in land cover and climate to generalize degradation patterns across districts.

Data & Methodology

Data Sources

Land Cover Fractions	Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland (8 classes per grid cell)
Climate & Soil Variables	NDVI, Rainfall, Temperature, Soil Moisture (averaged per grid cell)

Workflow

- Aggregate land cover fractions and climate/soil variables into 5 km × 5 km grid cells
- Compute a Land Degradation Index (LDI) per cell from these environmental indicators
- Label each cell's LDI into Low / Moderate / High degradation classes
- Train and tune a Logistic Regression classifier (13 environmental + 19 district features)
- Validate performance and interpret feature influence using SHAP explainability

Features Used in the Model

13 environmental variables + district identity feed the classifier

Land Cover Fractions (8)

Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland

Climate & Soil Variables (5)

NDVI_mean, Rainfall_mean, Temperature_mean, SoilMoisture_mean, Area_km2

District Identity

Each of the 20 Uttar Pradesh districts is one-hot encoded (19 binary features), letting the model capture district-specific baseline conditions.

Total input features per grid cell: 13 environmental + 19 district (one-hot) = 32

Degradation Severity Classification

Every grid cell is assigned one of three degradation classes

Class	Description
Low	Healthy vegetation cover, stable soil moisture, minimal bare land or built-up encroachment
Moderate	Noticeable vegetation loss or rising bare/built-up share; land under stress but recoverable
High	Severe vegetation loss, high bare land, low soil moisture — urgent restoration needed

Model used: Tuned Logistic Regression, trained on 13 environmental features plus district identity. It outputs the Low / Moderate / High class with a confidence score for every 5 km × 5 km grid cell.

Land Degradation Prediction System

Machine-learning powered monitoring for Uttar Pradesh, India (2020–2024)

Deployed Model

Model	Tuned Logistic Regression
Task	Multiclass classification
Classes	Low, Moderate, High
Features	13 environmental + 19 district (one-hot)

Artefact Status

- ✓ best_model — ready
- ✓ preprocessor_lr — ready
- ✓ label_encoder — ready

Scope

20 districts covered · 5 km × 5 km grid resolution · 22,645 grid observations · 3 severity classes

Platform Features

Six modules covering prediction, analysis, and interpretation

Single Prediction — Enter values for one grid cell and get an instant Low/Moderate/High prediction with confidence score

Batch Prediction — Upload a CSV of many grid cells at once — tested successfully on 22,645 rows in one run

Model Performance — Metrics and validation results for the Tuned Logistic Regression classifier

EDA Dashboard — Explore observations by year and class — distributions, correlations, and scatter analysis

SHAP Explainability — Shows which features (temperature, rainfall, soil moisture, etc.) drive each prediction

Geographic Dashboard — Maps degradation results spatially across Uttar Pradesh's districts

Sample Prediction in Action

Single Prediction example — Chandauli district

Sample Inputs

District	Chandauli
Area (km ²)	20.42
Cropland	73.84%
TreeCover	7.45%
BareLand	0.23%
Built-up	2.98%
NDVI (mean)	0.475
Rainfall (mean)	1051.5 mm
Temperature (mean)	25.44 °C
Soil Moisture (mean)	0.240

Prediction Result

Moderate

Confidence: 100.0% · Model: Tuned Logistic Regression

Class Probabilities

Low	0.0%
Moderate	100.0%
High	0.0%

This grid cell shows strong cropland dominance with moderate tree cover — flagged for monitoring rather than immediate intervention.

Conclusion & Future Scope

- A Tuned Logistic Regression model classifies land degradation (Low/Moderate/High) across 22,645 grid cells in 20 Uttar Pradesh districts

The live platform offers single & batch prediction, EDA, model performance, and SHAP explainability — all deployed and working

Future scope: extend coverage to more states, automate satellite data ingestion, and add mobile access for field officers

Goal: support local authorities and communities in prioritizing restoration efforts using data-driven insights

Thank You